

# Introduction to Statistics

## Statistics I • Section 1

Zong-Wei Yeh

Department of Finance, National Chung Cheng University

Fall 2026

# Outline

---

1.1 What Is Statistics?

1.2 Population and Sample

1.3 Data Categories and Data Issues



# **What Is Statistics?**

# Decisions are made without complete information

---

- A fund manager chooses a portfolio before knowing next year's returns. A firm launches a product without asking every buyer.
- We observe a **part**. We must reason about the **whole**.

## The chain

data → information → decision-making

- Data settle nothing until we summarize them. A summary guides action only after we know how far to trust it.

# Two modes of the subject

---

## Definition 1.1 (Descriptive and inferential statistics)

1. *Descriptive* (敘述統計): procedures that summarize a data set.
2. *Inferential* (推論統計): procedures that use a sample to draw conclusions about the population it came from.
  - Average return of *the 30 stocks you hold*: a \_\_\_\_\_ statement. It is exactly right.
  - Average of 30 random stocks, used to speak about a market of 900: an \_\_\_\_\_ statement. It **can** be wrong.
  - Inference always carries uncertainty. This course exists to measure it, not to hide it.

# Two modes of the subject

---

## Definition 1.1 (Descriptive and inferential statistics)

1. *Descriptive* (敘述統計): procedures that summarize a data set.
  2. *Inferential* (推論統計): procedures that use a sample to draw conclusions about the population it came from.
- Average return of *the 30 stocks you hold*: a descriptive statement. It is exactly right.
  - Average of 30 random stocks, used to speak about a market of 900: an \_\_\_\_\_ statement. It **can** be wrong.
  - Inference always carries uncertainty. This course exists to measure it, not to hide it.

# Two modes of the subject

---

## Definition 1.1 (Descriptive and inferential statistics)

1. *Descriptive* (敘述統計): procedures that summarize a data set.
  2. *Inferential* (推論統計): procedures that use a sample to draw conclusions about the population it came from.
- Average return of *the 30 stocks you hold*: a descriptive statement. It is exactly right.
  - Average of 30 random stocks, used to speak about a market of 900: an inferential statement. It **can** be wrong.
  - Inference always carries uncertainty. This course exists to measure it, not to hide it.



# Population and Sample

# Population and sample

---

## Definition 1.2 (Population and sample)

1. *Population* (母體): the \_\_\_\_\_ set of items of interest, size  $N$  (possibly infinite).
  2. *Sample* (樣本): an \_\_\_\_\_ of it, size  $n$ .
- The **question** fixes the population. The data you happen to have do not.
  - To study Taiwanese households, one brokerage's clients are **not** the population. Interviewing more of them does not fix this.
  - Account holders are already a special group. Households with no account never reach the list.

# Population and sample

---

## Definition 1.2 (Population and sample)

1. *Population* (母體): the complete set of items of interest, size  $N$  (possibly infinite).
  2. *Sample* (樣本): an \_\_\_\_\_ of it, size  $n$ .
- The **question** fixes the population. The data you happen to have do not.
  - To study Taiwanese households, one brokerage's clients are **not** the population. Interviewing more of them does not fix this.
  - Account holders are already a special group. Households with no account never reach the list.

# Population and sample

---

## Definition 1.2 (Population and sample)

1. *Population* (母體): the complete set of items of interest, size  $N$  (possibly infinite).
  2. *Sample* (樣本): an observed subset of it, size  $n$ .
- The **question** fixes the population. The data you happen to have do not.
  - To study Taiwanese households, one brokerage's clients are **not** the population. Interviewing more of them does not fix this.
  - Account holders are already a special group. Households with no account never reach the list.

# Why not just measure everyone?

---

- To know a population number for certain, measure every member. That is a **census** (普查).
- A census is rarely available. Three obstacles recur:
  - **cost and time**: Taiwan's census runs once every ten years, so no firm can repeat it to answer a question this month;
  - **measuring destroys the item**: if we test every light bulb to failure, we have nothing left to sell;
  - **the population cannot be listed at all**.
- So we **sample**: measure a part, reason about the whole.

## A population that does not exist yet

---

- A population need not be a set of things that already exist.
- Take the daily return on a stock. We may hold 250 trading days of data. Those 250 days are not what we care about.
- We care about the **mechanism** that produced them. The population is every return that mechanism could produce.
- It has no list. It is infinite, and most of it has not happened yet.
- Such a population is called \_\_\_\_\_ . No census can reach it. Sections 3–7 model it instead.

## A population that does not exist yet

---

- A population need not be a set of things that already exist.
- Take the daily return on a stock. We may hold 250 trading days of data. Those 250 days are not what we care about.
- We care about the **mechanism** that produced them. The population is every return that mechanism could produce.
- It has no list. It is infinite, and most of it has not happened yet.
- Such a population is called conceptual (概念性母體). No census can reach it. Sections 3–7 model it instead.

# Parameter vs. statistic: the asymmetry

---

## Definition 1.3 (Parameter and statistic)

1. *Parameter* (母數 / 參數): a numerical measure of a **population** characteristic. It is \_\_\_\_\_, and usually unknown.
  2. *Statistic* (統計量): a numerical measure computed from a **sample**. We know its value once we have the sample, but it \_\_\_\_\_.
- Parameter  $\mu$  (all 900 stocks): fixed, unknown. Statistic  $\bar{x}$  (your 30 stocks): known, but changes with each sample.
  - What we **want** is constant and hidden. What we **have** is visible and random.
  - Section 7 makes this precise: a statistic is a random variable.

# Parameter vs. statistic: the asymmetry

---

## Definition 1.3 (Parameter and statistic)

1. *Parameter* (母數 / 參數): a numerical measure of a **population** characteristic. It is fixed, and usually unknown.
  2. *Statistic* (統計量): a numerical measure computed from a **sample**. We know its value once we have the sample, but it \_\_\_\_\_.
- Parameter  $\mu$  (all 900 stocks): fixed, unknown. Statistic  $\bar{x}$  (your 30 stocks): known, but changes with each sample.
  - What we **want** is constant and hidden. What we **have** is visible and random.
  - Section 7 makes this precise: a statistic is a random variable.

# Parameter vs. statistic: the asymmetry

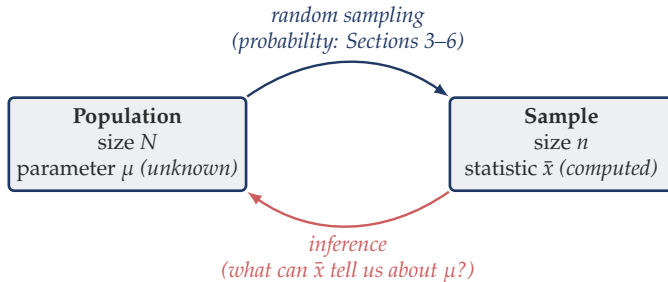
---

## Definition 1.3 (Parameter and statistic)

1. *Parameter* (母數 / 參數): a numerical measure of a **population** characteristic. It is fixed, and usually unknown.
  2. *Statistic* (統計量): a numerical measure computed from a **sample**. We know its value once we have the sample, but it differs from sample to sample.
- Parameter  $\mu$  (all 900 stocks): fixed, unknown. Statistic  $\bar{x}$  (your 30 stocks): known, but changes with each sample.
  - What we **want** is constant and hidden. What we **have** is visible and random.
  - Section 7 makes this precise: a statistic is a random variable.

# The inference loop

---



- We can model the forward arrow, because **we** choose the sampling mechanism. The backward arrow is only as good as that model.
- So the honest question is not “is  $\bar{x}$  right?” It is “how far off can it plausibly be?”

# How the sample must be drawn

---

## Definition 1.4 (Simple random sample)

A *simple random sample* (簡單隨機樣本): every member of the population is \_\_\_\_\_ to be chosen; no member's selection influences another's; every possible sample of size  $n$  is equally likely.

- It constrains the \_\_\_\_\_, not the result: a random sample can be unlucky, and we can compute how often that happens.
- Randomness does not guarantee a representative sample; it guarantees a **quantifiable** chance of an unrepresentative one.
- A convenience sample gives us nothing. We do not know its mechanism, so there is nothing to compute.

# How the sample must be drawn

---

## Definition 1.4 (Simple random sample)

A *simple random sample* (簡單隨機樣本): every member of the population is equally likely to be chosen; no member's selection influences another's; every possible sample of size  $n$  is equally likely.

- It constrains the \_\_\_\_\_, not the result: a random sample can be unlucky, and we can compute how often that happens.
- Randomness does not guarantee a representative sample; it guarantees a **quantifiable** chance of an unrepresentative one.
- A convenience sample gives us nothing. We do not know its mechanism, so there is nothing to compute.

# How the sample must be drawn

---

## Definition 1.4 (Simple random sample)

A *simple random sample* (簡單隨機樣本): every member of the population is equally likely to be chosen; no member's selection influences another's; every possible sample of size  $n$  is equally likely.

- It constrains the procedure, not the result: a random sample can be unlucky, and we can compute how often that happens.
- Randomness does not guarantee a representative sample; it guarantees a quantifiable chance of an unrepresentative one.
- A convenience sample gives us nothing. We do not know its mechanism, so there is nothing to compute.

## Systematic sampling (系統抽樣)

---

- Order the list, random start among the first  $j$ , then take every  $j$ th item, where

$$j = \frac{N}{n}.$$

- Cheap, and it behaves like a random sample. But that holds only **if** the ordering is unrelated to the variable we study.
- Danger: a \_\_\_\_\_ pattern of period  $j$  (or dividing  $j$ ). Every selected item then sits at the same point of the cycle.

## Systematic sampling (系統抽樣)

---

- Order the list, random start among the first  $j$ , then take every  $j$ th item, where

$$j = \frac{N}{n}.$$

- Cheap, and it behaves like a random sample. But that holds only **if** the ordering is unrelated to the variable we study.
- Danger: a \_\_\_\_\_ pattern of period  $j$  (or dividing  $j$ ). Every selected item then sits at the same point of the cycle.

## Systematic sampling (系統抽樣)

---

- Order the list, random start among the first  $j$ , then take every  $j$ th item, where

$$j = \frac{N}{n}.$$

- Cheap, and it behaves like a random sample. But that holds only **if** the ordering is unrelated to the variable we study.
- Danger: a **cyclical** pattern of period  $j$  (or dividing  $j$ ). Every selected item then sits at the same point of the cycle.

## Example 1.5

---

### Example 1.5

A brokerage's client list holds  $N = 8,000$  accounts, ordered by the date each was opened. An analyst wants a systematic sample of  $n = 200$ .

- (a) Find the sampling interval  $j$ .
- (b) With random start 7, give the first three accounts selected and the last one.
- (c) Describe one ordering that would make the sample misleading.

## Two kinds of error

---

- **Sampling error** (抽樣誤差): from observing  $n$  items, not all  $N$ . It is unavoidable, and it is \_\_\_\_\_. It shrinks as  $n$  grows.
- **Nonsampling error** (非抽樣誤差): it would remain even in a full \_\_\_\_\_.
  - **Wrong frame**: the sampled population is not the target;
  - **Inaccurate response**: answers that are ambiguous, leading or dishonest;
  - \_\_\_\_\_ : you sample those *willing to respond*.
- Sampling error shrinks toward 0 as  $n$  grows. A nonsampling *bias* does not. A biased survey of a million is no better than one of a thousand.

## Two kinds of error

---

- **Sampling error** (抽樣誤差): from observing  $n$  items, not all  $N$ . It is unavoidable, and it is quantifiable. It shrinks as  $n$  grows.
- **Nonsampling error** (非抽樣誤差): it would remain even in a full \_\_\_\_\_.
  - **Wrong frame**: the sampled population is not the target;
  - **Inaccurate response**: answers that are ambiguous, leading or dishonest;
  - \_\_\_\_\_ : you sample those *willing to respond*.
- Sampling error shrinks toward 0 as  $n$  grows. A nonsampling *bias* does not. A biased survey of a million is no better than one of a thousand.

## Two kinds of error

---

- **Sampling error** (抽樣誤差): from observing  $n$  items, not all  $N$ . It is unavoidable, and it is quantifiable. It shrinks as  $n$  grows.
- **Nonsampling error** (非抽樣誤差): it would remain even in a full census.
  - **Wrong frame**: the sampled population is not the target;
  - **Inaccurate response**: answers that are ambiguous, leading or dishonest;
  - \_\_\_\_\_ : you sample those *willing to respond*.
- Sampling error shrinks toward 0 as  $n$  grows. A nonsampling *bias* does not. A biased survey of a million is no better than one of a thousand.

## Two kinds of error

---

- **Sampling error** (抽樣誤差): from observing  $n$  items, not all  $N$ . It is unavoidable, and it is quantifiable. It shrinks as  $n$  grows.
- **Nonsampling error** (非抽樣誤差): it would remain even in a full census.
  - **Wrong frame**: the sampled population is not the target;
  - **Inaccurate response**: answers that are ambiguous, leading or dishonest;
  - **Nonresponse** (無回應): you sample those *willing to respond*.
- Sampling error shrinks toward 0 as  $n$  grows. A nonsampling *bias* does not. A biased survey of a million is no better than one of a thousand.

---

## **Data Categories and Data Issues**

# Kinds of variable

---

## Definition 1.6 (Variable; categorical and numerical)

1. *Categorical* (類別變數): values are groups or labels. This holds even when we store them as numeric codes.
  2. *Numerical* (數值變數): quantities from counting or measuring. It is *discrete* (離散) if its values can be listed one by one, *continuous* (連續) if it can take any value in an interval.
- Number of trades placed in a month: \_\_\_\_\_ (its values can be listed one by one).
  - Market value of a portfolio: \_\_\_\_\_ (any value in an interval).

# Kinds of variable

---

## Definition 1.6 (Variable; categorical and numerical)

1. *Categorical* (類別變數): values are groups or labels. This holds even when we store them as numeric codes.
  2. *Numerical* (數值變數): quantities from counting or measuring. It is *discrete* (離散) if its values can be listed one by one, *continuous* (連續) if it can take any value in an interval.
- Number of trades placed in a month: discrete (its values can be listed one by one).
  - Market value of a portfolio: \_\_\_\_\_ (any value in an interval).

# Kinds of variable

---

## Definition 1.6 (Variable; categorical and numerical)

1. *Categorical* (類別變數): values are groups or labels. This holds even when we store them as numeric codes.
  2. *Numerical* (數值變數): quantities from counting or measuring. It is *discrete* (離散) if its values can be listed one by one, *continuous* (連續) if it can take any value in an interval.
- Number of trades placed in a month: discrete (its values can be listed one by one).
  - Market value of a portfolio: continuous (any value in an interval).

## The trap: categories coded as numbers

---

- Satisfaction stored as  $1, \dots, 5$ ; gender as  $1, 2$ . The entries **are** numerals.
- That does not make the variable numerical. “5” is a label for “very satisfied”, fixed by convention. It could be “A”.
- A coded category is still a \_\_\_\_\_ variable.
- Mirror trap: rounding a share price to two decimals does not make it discrete. The quantity still varies over a continuum.

## The trap: categories coded as numbers

---

- Satisfaction stored as  $1, \dots, 5$ ; gender as  $1, 2$ . The entries **are** numerals.
- That does not make the variable numerical. “5” is a label for “very satisfied”, fixed by convention. It could be “A”.
- A coded category is still a categorical variable.
- Mirror trap: rounding a share price to two decimals does not make it discrete. The quantity still varies over a continuum.

# Measurement levels

---

## Definition 1.7 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
  2. *Ordinal* (順序尺度): we can \_\_\_\_ the values, but no fixed differences.
  3. *Interval* (等距尺度): differences meaningful, zero \_\_\_\_\_.
  4. *Ratio* (比率尺度): differences meaningful, zero \_\_\_\_\_.
- 
- Nominal/ordinal are the levels of categorical variables; interval/ratio are the levels of \_\_\_\_\_ ones.
  - Separating the last two: is the zero a \_\_\_\_\_ (20°C not twice 10°C) or genuine (NT\$2m twice NT\$1m)?

# Measurement levels

---

## Definition 1.7 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
  2. *Ordinal* (順序尺度): we can rank the values, but no fixed differences.
  3. *Interval* (等距尺度): differences meaningful, zero \_\_\_\_\_.
  4. *Ratio* (比率尺度): differences meaningful, zero \_\_\_\_\_.
- 
- Nominal/ordinal are the levels of categorical variables; interval/ratio are the levels of \_\_\_\_\_ ones.
  - Separating the last two: is the zero a \_\_\_\_\_ (20°C not twice 10°C) or genuine (NT\$2m twice NT\$1m)?

# Measurement levels

---

## Definition 1.7 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
  2. *Ordinal* (順序尺度): we can rank the values, but no fixed differences.
  3. *Interval* (等距尺度): differences meaningful, zero arbitrary.
  4. *Ratio* (比率尺度): differences meaningful, zero \_\_\_\_\_.
- Nominal/ordinal are the levels of categorical variables; interval/ratio are the levels of \_\_\_\_\_ ones.
  - Separating the last two: is the zero a \_\_\_\_\_ ( $20^{\circ}\text{C}$  not twice  $10^{\circ}\text{C}$ ) or genuine (NT\$2m twice NT\$1m)?

# Measurement levels

---

## Definition 1.7 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
  2. *Ordinal* (順序尺度): we can rank the values, but no fixed differences.
  3. *Interval* (等距尺度): differences meaningful, zero arbitrary.
  4. *Ratio* (比率尺度): differences meaningful, zero genuine.
- 
- Nominal/ordinal are the levels of categorical variables; interval/ratio are the levels of \_\_\_\_\_ ones.
  - Separating the last two: is the zero a \_\_\_\_\_ (20°C not twice 10°C) or genuine (NT\$2m twice NT\$1m)?

# Measurement levels

---

## Definition 1.7 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
  2. *Ordinal* (順序尺度): we can rank the values, but no fixed differences.
  3. *Interval* (等距尺度): differences meaningful, zero arbitrary.
  4. *Ratio* (比率尺度): differences meaningful, zero genuine.
- Nominal/ordinal are the levels of categorical variables; interval/ratio are the levels of numerical ones.
  - Separating the last two: is the zero a \_\_\_\_\_ (20°C not twice 10°C) or genuine (NT\$2m twice NT\$1m)?

# Measurement levels

---

## Definition 1.7 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
  2. *Ordinal* (順序尺度): we can rank the values, but no fixed differences.
  3. *Interval* (等距尺度): differences meaningful, zero arbitrary.
  4. *Ratio* (比率尺度): differences meaningful, zero genuine.
- Nominal/ordinal are the levels of categorical variables; interval/ratio are the levels of numerical ones.
  - Separating the last two: is the zero a convention (20°C not twice 10°C) or genuine (NT\$2m twice NT\$1m)?

## What each level licenses

---

| Level    | Supports            | Does not support     | Example          |
|----------|---------------------|----------------------|------------------|
| Nominal  | counting, mode      | ranking, differences | industry sector  |
| Ordinal  | ranking, median     | differences, means   | bond rating      |
| Interval | differences, mean   | ratios               | temperature (°C) |
| Ratio    | differences, ratios | —                    | portfolio value  |

- Cumulative downward: each row supports everything the row above does, and one thing more.
- Averaging a 1–5 scale: codes carry order, not \_\_\_\_\_, so the mean has no calibrated meaning. The \_\_\_\_\_ and the frequency distribution are defensible.

## What each level licenses

---

| Level    | Supports            | Does not support     | Example          |
|----------|---------------------|----------------------|------------------|
| Nominal  | counting, mode      | ranking, differences | industry sector  |
| Ordinal  | ranking, median     | differences, means   | bond rating      |
| Interval | differences, mean   | ratios               | temperature (°C) |
| Ratio    | differences, ratios | —                    | portfolio value  |

- Cumulative downward: each row supports everything the row above does, and one thing more.
- Averaging a 1–5 scale: codes carry order, not distance, so the mean has no calibrated meaning. The \_\_\_\_\_ and the frequency distribution are defensible.

## What each level licenses

---

| Level    | Supports            | Does not support     | Example          |
|----------|---------------------|----------------------|------------------|
| Nominal  | counting, mode      | ranking, differences | industry sector  |
| Ordinal  | ranking, median     | differences, means   | bond rating      |
| Interval | differences, mean   | ratios               | temperature (°C) |
| Ratio    | differences, ratios | —                    | portfolio value  |

- Cumulative downward: each row supports everything the row above does, and one thing more.
- Averaging a 1–5 scale: codes carry order, not distance, so the mean has no calibrated meaning. The median and the frequency distribution are defensible.

## Example 1.8

---

### Example 1.8

A bank stores three fields for each of its 500 corporate clients: the credit rating (AAA, AA, ..., C); the calendar year the account was opened; the balance in NT\$. For each computation, say whether it is meaningful, and why.

- (a) The mean credit rating.
- (b) The difference between two clients' opening years.
- (c) The ratio of two clients' balances.
- (d) The ratio of two clients' opening years.

## Selection bias (選擇偏誤)

---

- The sample admits items for a reason tied to the variable we study. The sampled population then differs from the target.
- A brokerage estimates the average return of **all** retail investors from its *premium-research subscribers*, who trade more and hold more.
- The \_\_\_\_\_ it sampled is not the target population; the sample can be huge and the bias survives.
- Everyday version, \_\_\_\_\_ : online reviews are written by customers who chose to write one; a fund index built on voluntary reporting describes funds *willing to report*.

## Selection bias (選擇偏誤)

---

- The sample admits items for a reason tied to the variable we study. The sampled population then differs from the target.
- A brokerage estimates the average return of **all** retail investors from its *premium-research subscribers*, who trade more and hold more.
- The frame it sampled is not the target population; the sample can be huge and the bias survives.
- Everyday version, \_\_\_\_\_ : online reviews are written by customers who chose to write one; a fund index built on voluntary reporting describes funds *willing to report*.

## Selection bias (選擇偏誤)

---

- The sample admits items for a reason tied to the variable we study. The sampled population then differs from the target.
- A brokerage estimates the average return of **all** retail investors from its *premium-research subscribers*, who trade more and hold more.
- The frame it sampled is not the target population; the sample can be huge and the bias survives.
- Everyday version, self-selection (自我選擇): online reviews are written by customers who chose to write one; a fund index built on voluntary reporting describes funds *willing to report*.

## Bias in the time dimension

---

- Selection bias also works in the **time** dimension. This version matters most for return data.
- A fund database holds the funds that *still exist*: funds that did badly were closed or merged, taking their returns with them.
- Averaging over the survivors measures a group selected, after the fact, for having done well. The data show no sign of this. We call it  
\_\_\_\_\_ bias.
- No larger sample helps: a bigger sample of survivors is still a sample of survivors.

## Bias in the time dimension

---

- Selection bias also works in the **time** dimension. This version matters most for return data.
- A fund database holds the funds that *still exist*: funds that did badly were closed or merged, taking their returns with them.
- Averaging over the survivors measures a group selected, after the fact, for having done well. The data show no sign of this. We call it survivorship (存活者偏誤) bias.
- No larger sample helps: a bigger sample of survivors is still a sample of survivors.

## Example 1.9

---

### Example 1.9

Ten years ago a database tracked 100 equity funds; today 70 survive and 30 were closed or merged. Over the decade the survivors averaged 8.0% per year; the 30 that disappeared averaged 2.0% per year while they existed. A magazine reports: “the average equity fund returned 8.0% a year.”

- (a) Equal-weighting all 100 original funds, what did the cohort average?
- (b) By how much does the magazine overstate it?
- (c) Which population does 8.0% describe, and why is it useless to an investor today?

# Takeaways

---

1. Descriptive statistics summarize. Inferential statistics reason from a sample to a population, and never with certainty.
2. Parameter (fixed, unknown) vs. statistic (known, random): the asymmetry the course is built on.
3. Sampling error shrinks as  $n$  grows. Nonsampling bias does not.
4. The measurement level decides which summaries are meaningful.
5. Selection and survivorship bias corrupt a data set however large. Always ask which population the number describes.